

DPPL-xx

DESKRIPSI PERANCANGAN PERANGKAT LUNAK

INTERLECT

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Group: Underrated

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	Prodi S1- Teknologi Informasi Universitas Telkom	Nomor Dokumen		Halaman
		<i>DPPL-xx</i> <xx:no grp>		<#>/<jml #
		Revisi	<nomor revisi>	Tgl: <isi tanggal>

LIST OF CHANGES

Revisi	Deskripsi
A	Revised Use Case Scenario from table to template-based.
B	Revised Class Diagram from standard to MVC Model Class Diagram.
C	Adjusted the Class Diagram table so it's based on the MVC Model Class Diagram.
D	
E	
F	
G	

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Ditulis oleh								
Diperiksa oleh								
Disetujui oleh								

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1. Introduction

1.1 Purpose of Writing the Document

This DPPL document serves as a comprehensive design specification for the Interlect application. It acts as a guide for system developers, guaranteeing that all functional and non-functional requirements listed in the SRS are appropriately converted into technical components.

1.2 Scope of the Problem

This documentation covers the specification for the Interlect application, which is designed to help students find their desired international education/college. The application's core functionality includes finding colleges based on a user's interest, current budget, and capability, as well as providing live consulting and AI-based education finding features. The system aims to inform students of their desired college and the necessary administration procedures.

1.3 Definitions and Terms

Term/Acronym	Definition
Interlect	The International Education Finder software/application.
SRS	Software Requirements Specification.
SKPL	Spesifikasi Kebutuhan Perangkat Lunak (Indonesian for Software Requirements Specification).
AI	Artificial Intelligence.
GPA	Grade Point Average.
IELTS	International English Language Testing System.
User	Refers to the Student user class.
Consultant	Refers to the Expert/Administrator user class.

1.4 References

-[1] Software Requirements Specification (SKPL) for Interlect, Group "underrated", 2025.

2 Global Design

2.1 Implementation Environment Plan

The **Interlect** system will be implemented using a **Client-Server Architecture** to ensure scalability and accessibility for students and consultants .

1. Client-Side (Front-end):

1.1 Platforms: Mobile Application (Android/iOS) and Web Browser.

1.2 Interface: Designed according to the UI specifications (P001 - P00Y) , enabling users to access features like University Search and Live Chat.

2. Server-Side (Back-end):

2.1 Architecture Pattern: MVC (**Model-View-Controller**) as specified in the detailed design . This separates the user interface (View), business logic (Controller), and data structure (Model/Entity).

2.2 API: RESTful API to handle communication between the Client and the Database.

3. Database:

3.1 Type: Relational Database Management System (RDBMS) (e.g., MySQL or PostgreSQL) to store entities such as Student, Consultant, University, and ChatHistory .

4. External Integrations:

4.1 AI Service: Integration for the **AI Chatbot** feature to process natural language queries.

2.2 Software Architecture Description

No	Component Name	Description
1	AuthComponent	Manages user access, including Login (UserCredentialValidator), Sign Up, and Session Management .
2	SearchEngineComponent	Handles the logic for SearchController, processing filters (Budget, Country, GPA) to match Students with Universities .

3	CommunicationComponent	Manages ChatController for live sessions between Students and Consultants, including message logging (ChatHistory) .
4	ReviewComponent	Manages ReviewController to handle rating inputs and calculate average scores for universities .
5	AI_IntegrationComponent	Manages the interaction between the user and the AI Knowledge Base for the "Consult with AI" feature .
6	DatabaseLayer	Represents the persistent data entities (Model) such as User, University, and Data .

3 Detailed Design

3.1 Use Case Realization

Berisi TABEL USE CASE sebagai berikut :

No	Nama Use Case	Deskripsi Use Case
#1	Login	User/Admin can log in using their own account
#2		
#3	Live chat consulting session with professionals	User selects a consultant and engages in a consulting session.
#4	Give Review and Rating	User can give or check the review and rating score on every university.
#5	Consult with AI Chatbot	User can have a consultation with chatbot that already included in the software

3.1.1 Use Case #1 <Login>

3.1.1.1 Use Case Scenario #1 <Login>

Skenario Use Case #1 :

- i. Pre-Condition : Login
- ii. Use Case Description : The User or Admin already has an existing account.
 - a. Primary Flow :
 1. **User** initiates the login process.
 2. **User** sees the Login Form displayed on the screen.
 3. **User** enters their registered Email and Password.
 4. **User** clicks the "Login" button.
 5. **User** waits briefly while the system validates the credentials.
 6. **User** is successfully redirected and views the Home Page.
 - b. Alternative Flow :
 1. **User** starts the login process but realizes they don't have an account.
 2. **User** sees the Login Form and clicks on the "Sign Up" option.
 3. **User** views the new Sign-Up form.
 4. **User** chooses to register manually or via a third-party account.
 5. **User** fills in their Email, Username, and Password.
 6. **User** clicks the "Sign Up" button.
 7. **User** receives a confirmation that their data is saved and is granted immediate access to the system.
- iii. Post-Condition : The user is authenticated and has full access to their dashboard on the home screen.

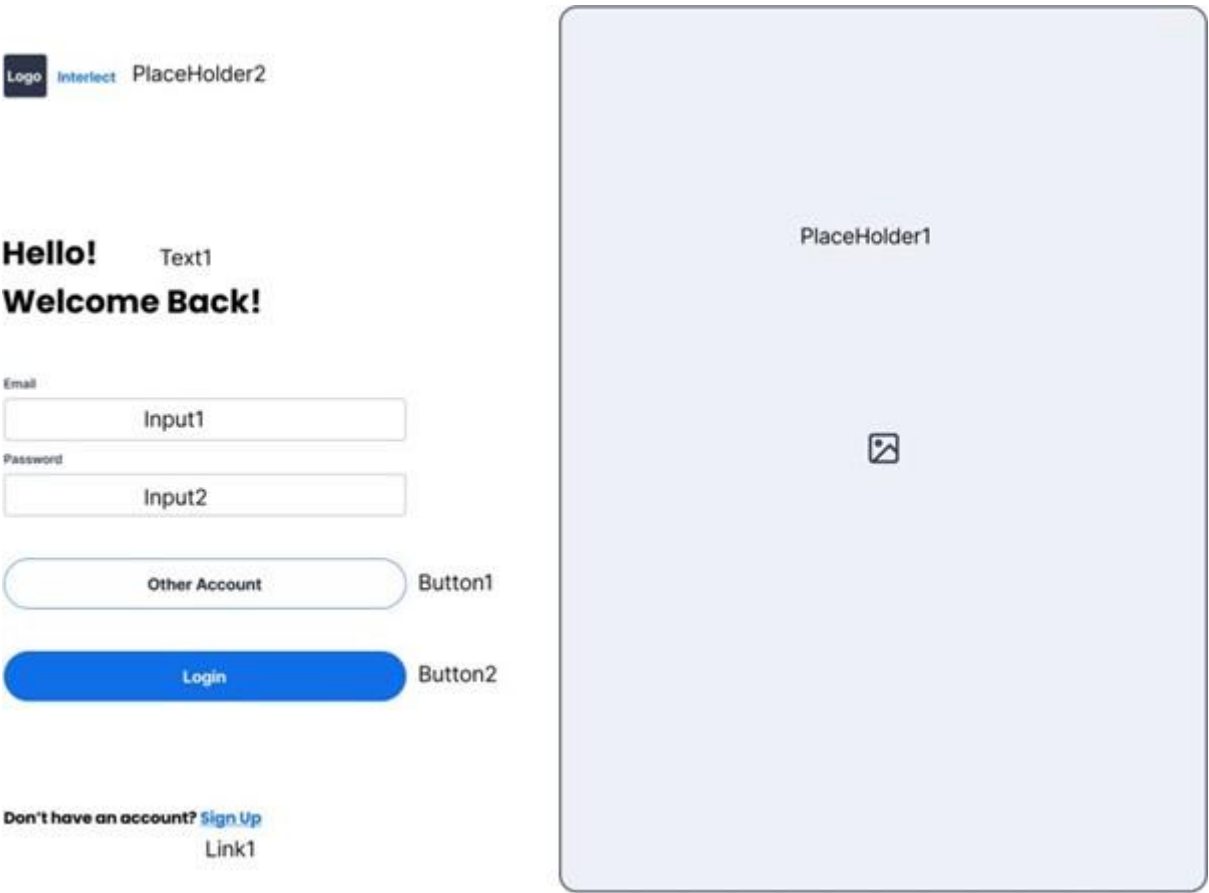
3.1.1.2 UI Design dan Deskripsi Objek UI #1 <Login>

Bagian ini diisi dengan versi awal prototipe antarmuka untuk per Use Case Selanjutnya, untuk setiap antarmuka/layar, tuliskan spesifikasi detilnya.

Tabel Deskripsi Objek UI

ID. PAGE	PAGE NAME	DESCRIPTION
P001	Login Page	<i>Authentication/Sign In Page. Allows users to input their Email and Password to access the system. Includes a Login Button, Other Account Button, and a link to the Sign Up Page.</i>

P002	Sign Up Page	User Registration Page. Allows new users to create an account by inputting Email, Password, and Confirm Password. Includes a Sign Up Button, Other Account Button, and a link back to the Login Page.
P003	Home Page	Main Application Landing Page after login (or the primary public-facing page). Features a header/navigation bar with a Logo, multiple Tabs, and two Buttons. The main area below the header is a content block/dashboard followed by a section with three content cards or sections.



P001: Page Login

OBJECT_ID	TYPE	LABEL*	DESCRIPTION
Input1	Input Field	Email	When clicked, the user will be able to input their

			<i>email</i>
<i>Input2</i>	<i>Input Field</i>	<i>Password</i>	<i>When clicked, the user will be able to input their email and password</i>
<i>PlaceHolder1</i>	<i>Image</i>	<i>Image</i>	<i>Placeholder for an image showing on the login page</i>
<i>PlaceHolder2</i>	<i>Logo</i>	<i>Logo Interlect</i>	<i>Placeholder for brand logo showing on the login page</i>
<i>Button1</i>	<i>Button</i>	<i>Other Account</i>	<i>When clicked, the system verifies credentials using other available accounts (i.e, Google account)</i>
<i>Button2</i>	<i>Button</i>	<i>Login</i>	<i>When clicked, the system verifies the account, and if successful, directs the user to the home page.</i>
<i>Text1</i>	<i>Header Text</i>	<i>Hello! Welcome Back!</i>	<i>Welcoming text in the login page</i>
<i>Link1</i>	<i>Link Text</i>	<i>Sign Up</i>	<i>Link to be clicked by the user if they don't have an account, will direct user to the Sign Up Page(P002).</i>


Placeholder4

Let's Get Started! Text2

Email

Password

Confirm Password

Other Account
Button3

Sign Up
Button4

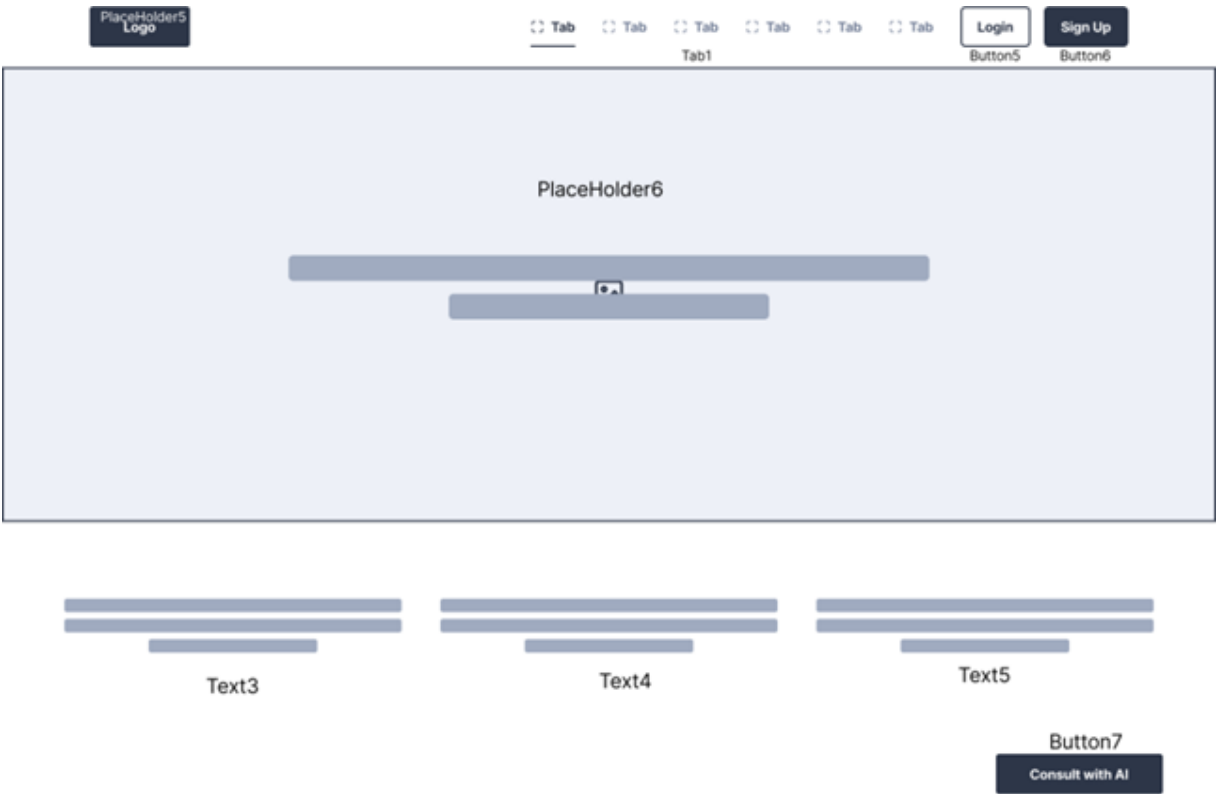
Already have an account? [Login](#) [Link2](#)

Placeholder3


P002: Page Sign Up

OBJECT_ID	TYPE	LABEL*	DESCRIPTION**
Input3	Input Field	Email	When clicked, the user will be able to input their email
Input4	Input Field	Password	When clicked, the user will be able to input their desired password
Input5	Input Field	Confirm Password	When clicked, the user will be able to input their confirmation of the desired password
Placeholder3	Image	Image	Placeholder for an image showing on the login page
Placeholder4	Image	Logo Interlect	Placeholder for brand logo showing on the login page
Button3	Button	Other Account	When clicked, the system verifies credentials using

			other available accounts (i.e, Google account), then continues to the home page
Button4	Button	Sign Up	When clicked, the system verifies the account, and if successful, directs the user to the home page.
Text2	Header Text	Let's Get Started!	Welcoming text in the Sign Up page.
Link2	Link Text	Login	Link to be clicked by the user if they already have an account, will direct the user to the Login page(P001).



P003: Home Page

OBJECT_ID	TYPE	LABEL*	DESCRIPTION**
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<i>Placeholder5</i>	<i>Image</i>	<i>Logo</i>	<i>Placeholder for brand logo showing on the home page</i>
<i>Placeholder6</i>	<i>Content Area</i>	<i>Main Content Header</i>	<i>Placeholder for the primary headline and sub-text (2 Lines Rectangle) in the main content area.</i>
<i>Text3</i>	<i>Text</i>	<i>Feature card 1</i>	<i>Placeholder to highlight one of the 3 features available</i>
<i>Text4</i>	<i>Text</i>	<i>Feature card 2</i>	<i>Placeholder to highlight the second feature</i>
<i>Text5</i>	<i>Text</i>	<i>Feature card 3</i>	<i>Placeholder to highlight the third feature</i>
<i>Tab1</i>	<i>Tab Menu</i>	<i>Tab 1-6</i>	<i>When clicked, the user will be directed to other content pages available</i>
<i>Button5</i>	<i>Button</i>	<i>Login</i>	<i>When clicked, the user will be directed to the login page(P001).</i>
<i>Button6</i>	<i>Button</i>	<i>Sign Up</i>	<i>When clicked, the user will be directed to the Sign Up page(P002).</i>
<i>Button7</i>	<i>Button</i>	<i>Consult with AI</i>	<i>When clicked, the user will be directed to the consultation with AI page</i>

3.1.1.3 Identifikasi Object dan Tipe Kelas #1 <Login>

OBJECT TABLE: LOGIN

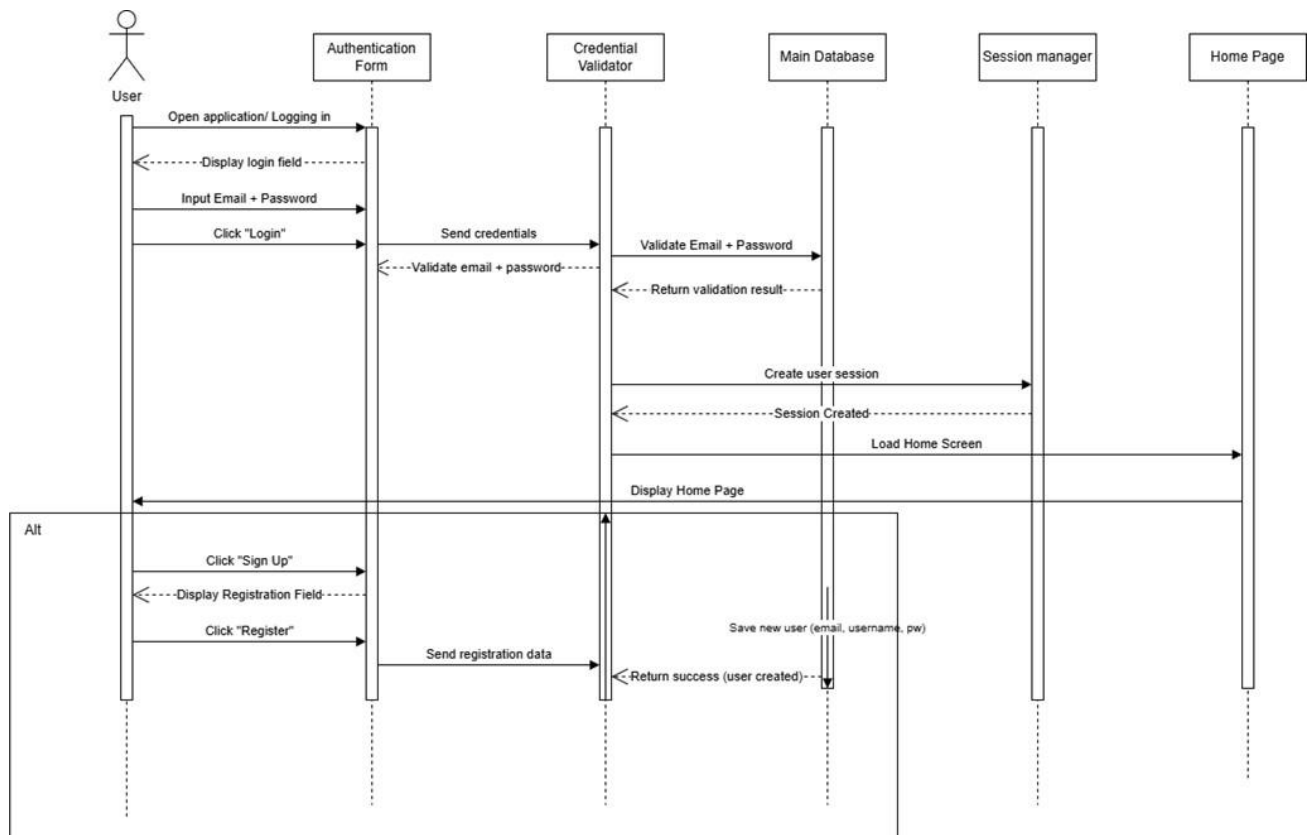
No	Object Name	Object Type	Description
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1	<i>Login Form</i>	<i>Boundary (Interface)</i>	<i>The user interface for entering (P001).</i>
2	<i>Registration Form</i>	<i>Boundary (Interface)</i>	<i>The user interface for creating a new account (P002).</i>
3	<i>Home Page</i>	<i>Boundary (Interface)</i>	<i>The main view is accessible after successful login or publicly (P003).</i>
4	<i>User</i>	<i>Entity (Database)</i>	<i>The object that stores user data and credentials (Name, Email Password)</i>
5	<i>Credential Validator</i>	<i>Controller (Processing)</i>	<i>The object responsible for processing and verifying login/registration data.</i>
6	<i>Session Manager</i>	<i>Controller (Processing)</i>	<i>The object responsible for creating, maintaining, and terminating a user's session.</i>
7	<i>Main database</i>	<i>Entity (Database)</i>	<i>The object that stores all persistent data for the system.</i>
8	<i>Feature</i>	<i>Entity (Database)</i>	<i>The object that stores details of each feature or service offered (as seen in the Feature cards on P003).</i>
9	<i>View Manager</i>	<i>Controller (Processing)</i>	<i>The object that determines and loads the correct content based on the user's login status and selected tab (based on Tabs in P003)._</i>
10	<i>Notification</i>	<i>Entity (Database)</i>	<i>The object that stores system messages or alerts for the user.</i>

**Tipe kelas seperti Boundary(Interface), Entity(Database), Controller*

3.1.1.4 Sequence Diagram #1 <Login>

Buatlah diagram sequence untuk masing - masing use case sesuai skenario.
Skenario harus melibatkan kelas-kelas perancangan yang baru diidentifikasi.



3.1.2 Use Case #2 <Search/Generate Universities>

3.1.2.1 Use Case Scenario #2 <Search/Generate Universities>

Skenario Use Case #2 :

- i. Pre-Condition : The user has already provided or fulfilled the necessary initial data required by the system.
- ii. Use Case Description : The user can generate a list of international schools or colleges tailored to their specific inputs, such as interest, budget, and academic capability.
 - a. Primary Flow :
 1. **User** clicks the "Generate University" button to start the process.
 2. **User** sees a new page containing the fields and data that need to be filled in.
 3. **User** inputs their personal data based on their specific interests, budget, and academic capability.
 4. **User** clicks the "Generate" button.
 5. **User** waits while the system generates a list of universities based on the provided input.
 6. **User** views the list and chooses to compare universities or view specific details for more information.
 7. **User** receives detailed information about the selected universities, whether for comparison or for reading further details.
 - b. Alternative Flow
 1. **User** enters criteria that do not match any existing records.

2. **User** sees an apologizing message stating that the system cannot find the desired university based on those inputs.
- iii. **Post-Condition** : The system successfully generates and displays matching universities based on the user's preferences.

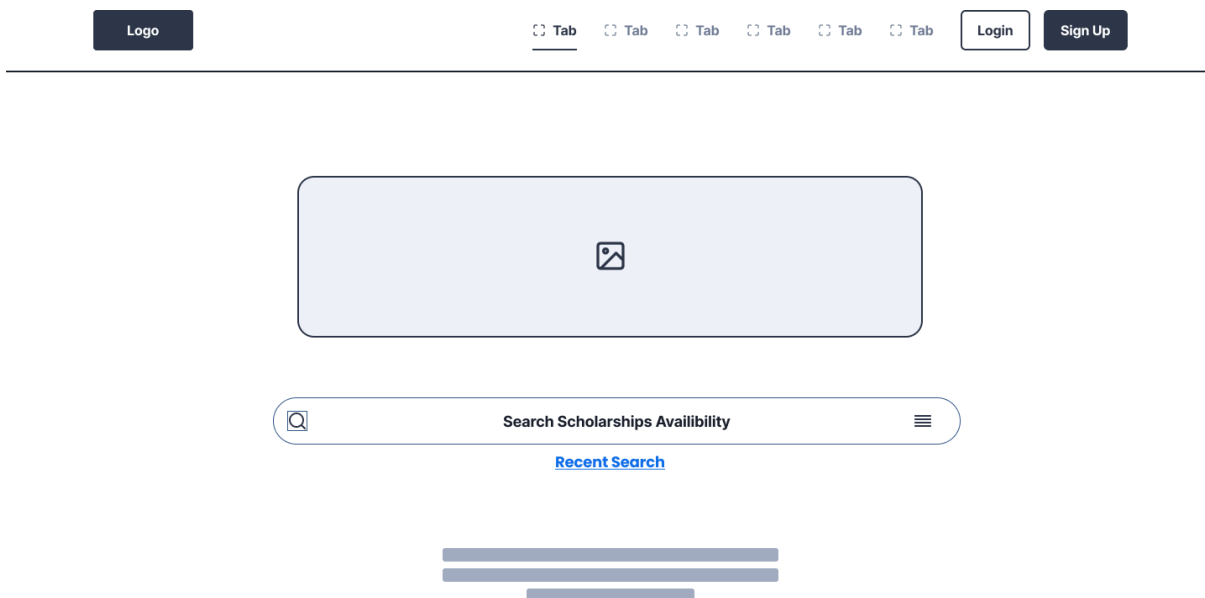
3.1.2.2 UI Design dan Deskripsi Objek UI #2 <Search/Generate Universities>

Bagian ini diisi dengan versi awal prototipe antarmuka untuk per Use Case
Selanjutnya, untuk setiap antarmuka/layar, tuliskan spesifikasi detilnya.

Tabel Deskripsi Objek UI

ID. LAYAR	NAMA LAYAR	DESKRIPSI
P001	Search Scholarship Availability	Main page for scholarship searching
P002	Filter Scholarships Panel	Scholarships filtering panel
P003	University Comparison Page	Page for comparing universities

UNTUK MASING – MASING ANTAR MUKA / PAGE dibuatkan spesifikasi detil
Antarmuka XXX: {diisi dengan no. layar atau no gambar rancangan antarmuka}



P001: Search Scholarship Availability

Id_Objek	JENIS	LABEL*	Keterangan**
TxtSearch	Text Field	Search Scholarships Availability	Input keyword for scholarship searching
IconSearch	Button	Search	Execute scholarship process(enter)

Id_Objek	JENIS	LABEL*	Keterangan**
<i>IconFilter</i>	<i>Button</i>	<i>Filter</i>	<i>Open filter panel</i>
<i>LnkRecent</i>	<i>Link</i>	<i>Recent search</i>	<i>Display search history</i>
<i>ResultList</i>	<i>Panel</i>	<i>-</i>	<i>Display scholarship results</i>

Logo

Tab

Tab

Tab

Tab

Tab

Tab

Login

Sign Up

Label

Label

Select

Large

Label

☐ Value

☐ Value

☐ Value

☐ Value

☐ Value

☐ Value

Search Scholarships Availability

Recent Search

P002: Filter Scholarships Panel

Id_Objek	JENIS	LABEL*	Keterangan**
<i>TxTKeyword</i>	<i>Text Field</i>	<i>Label</i>	<i>Additional keyword input</i>
<i>DdlCountry</i>	<i>Dropdown</i>	<i>Select Country</i>	<i>Choose target country</i>
<i>CbType</i>	<i>Checkbox</i>	<i>Value</i>	<i>Select scholarship type</i>
<i>BtnApply</i>	<i>Button</i>	<i>Apply</i>	<i>Applies filter criteria</i>

Id_Objek	JENIS	LABEL*	Keterangan**
<i>CardUni1</i>	<i>Card</i>	-	First university comparison card
<i>CardUni2</i>	<i>Card</i>	-	Secon university comparison card
<i>CardUni3</i>	<i>Card</i>	-	Third university for comparison
<i>DdlUni</i>	<i>Dropdwon</i>	Select	Select university information
<i>DescUni</i>	<i>Text Panel</i>	-	Displays university information

Identifikasi object yang terkait dengan use case tersebut.
Kelas pada tahap perancangan berbeda dengan kelas pada tahap analisis.
Gunakan tabel di bawah:

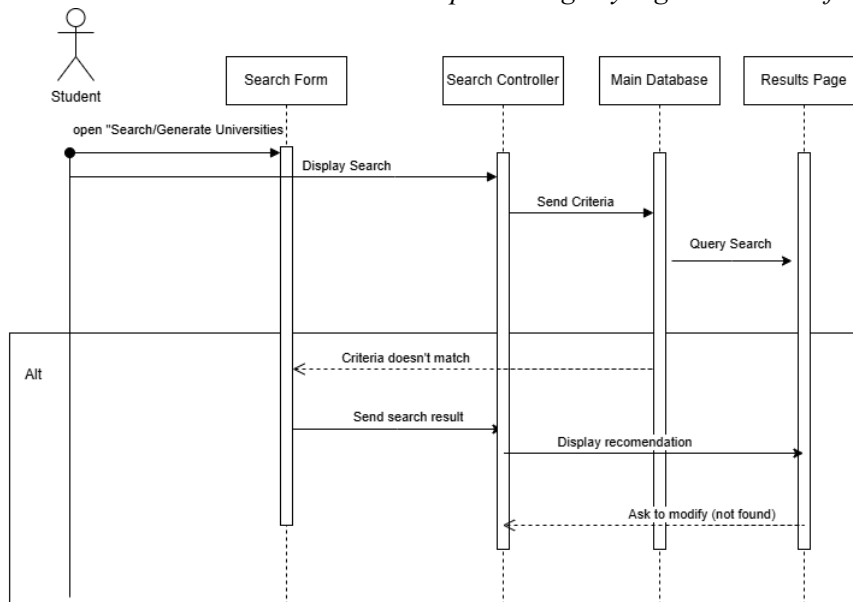
TABEL OBJECT PERANCANGAN

No	Nama Object	Jenis / Tipe Object
1	SearchScholarshipsPage	Boundary
2	ScholarshipSearchForm	Boundary
3	FilterScholarshipsPanel	Boundary
4	UniversityComparisonPage	Boundary
5	UniversityComparisonCard	Boundary
6	Scholarship	Entity
7	SearchHistory	Entity
8	FilterCriteria	Entity
9	University	Entity
10	ScholarshipSearchController	Controller
11	FilterController	Controller

**Tipe kelas seperti Boundary(Interface), Entity(Database), Controller*

3.1.2.4 Sequence Diagram #2 <Search/Generate Universities>

Buatlah diagram sequence untuk masing - masing use case sesuai skenario. Skenario harus melibatkan kelas-kelas perancangan yang baru diidentifikasi.



3.1.3 Use Case #3 <Live chat consulting session with professionals>

3.1.3.1 Use Case Scenario #3 <Live chat consulting session with professionals>

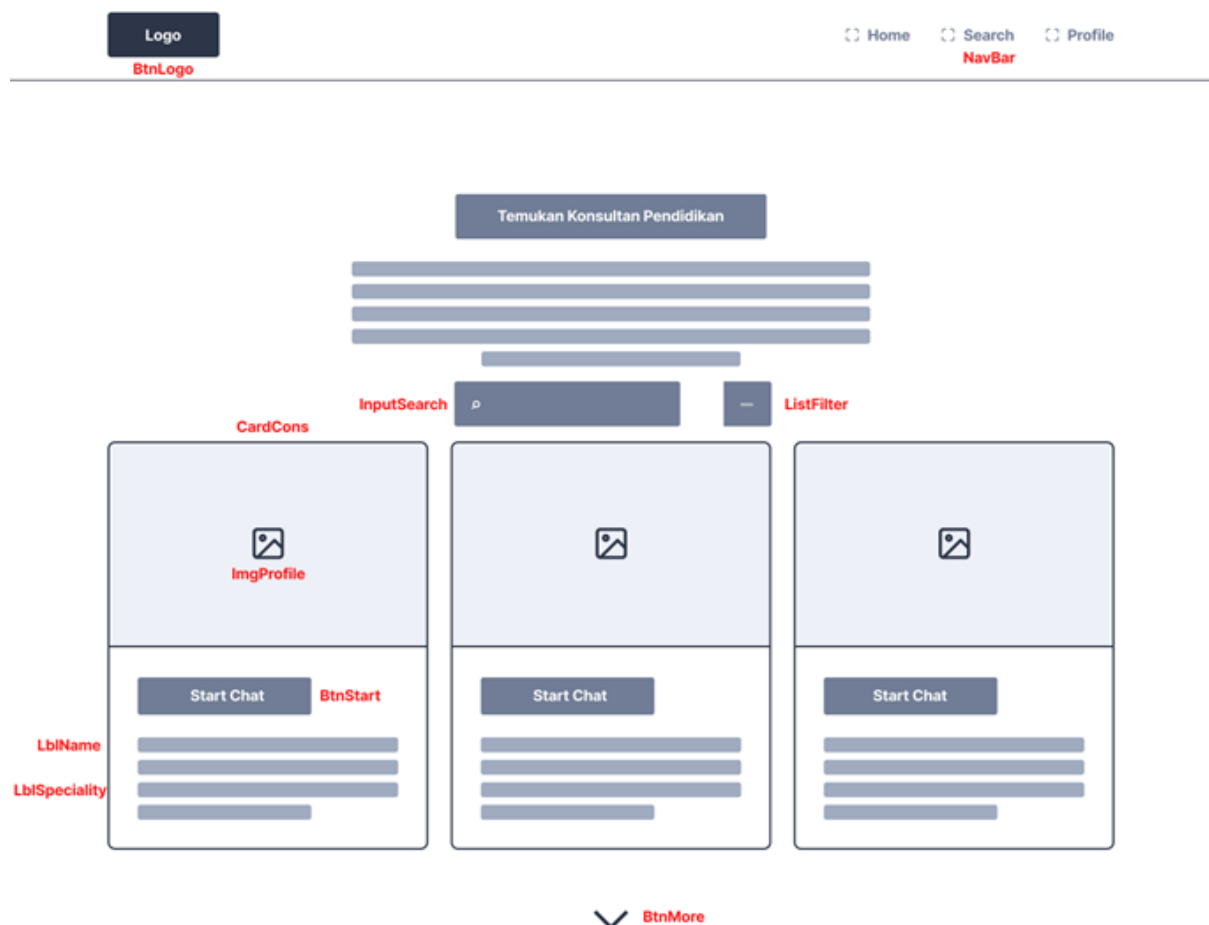
- i. Pre-Condition : User is logged in and is currently on Page List Consultant (P00X).
- ii. Use Case Description : Users can start a real-time consultation session with an available professional (Consultant) to get advice.
 - a. Primary Flow
 1. **User** inputs keywords into InputSearch or selects criteria in ListFilter on P00X.
 2. **User** processes the criteria via SearchController and CardCons will be updated to show matching consultants
 3. **User** clicks BtnStart inside a specific consultant's card
 4. **User** will be redirected to Page Chat Room (P00Y)
 5. **User** types a message into InputMessage and clicks BtnSend.
 6. **User** sees the message via ChatController,
 7. **User** clicks BtnEnd when the consultation is complete

8. **User** terminates the session, User will be redirected to the Rating Page or back to P00X.
- b. Alternative Flow
 1. **User** clicks BtnReturn instead of sending/ending..
 2. **User** sees the current session state paused..
 3. **User** will be redirected back to Page List Consultant (P00X) while keeping the session active in the background.
- iii. Post-Condition : The user knows what to expect from the desired university based on existing feedback from others.

3.1.3.2 UI Design and UI Object Description #3 <Live chat consulting session with professionals>

ID. PAGE	PAGE NAME	DESCRIPTION
P00X	Page List Consultant	The page displaying the search filters and a grid of consultant cards available for selection.
P00Y	Page Chat Room	The live interaction interface containing the chat area, action buttons (Return/End), and attachment tools.

Page LIST CONSULTANT / P00X



<i>Object_ID</i>	<i>TYPE</i>	<i>LABEL*</i>	<i>Description**</i>
<i>BtnLogo</i>	<i>Button/Image</i>	<i>Logo</i>	<i>The application logo. Clicking it returns the user to the Home Page.</i>
<i>NavBar</i>	<i>Navigation</i>	<i>Home, Search, Profile</i>	<i>Top navigation menu links.</i>
<i>InputSearch</i>	<i>Input Field</i>	<i>(Dynamic Content)</i>	<i>A card container holding the consultant's photo, name, and action button.</i>
<i>ListFilter</i>	<i>Container</i>	<i>(Filter Categories)</i>	<i>Container for category tags/dropdowns to sort the list.</i>
<i>CardCons</i>	<i>Container</i>	<i>(Dynamic Content)</i>	<i>A repeating card container holding the consultant's data.</i>
<i>ImgProfile</i>	<i>Image</i>	<i>(Photo)</i>	<i>Displays the consultant's profile picture inside the card.</i>
<i>LblName</i>	<i>Label</i>	<i>(Consultant Name)</i>	<i>Text inside the card displaying the consultant's name.</i>
<i>LblSpeciality</i>	<i>Label</i>	<i>(Description)</i>	<i>Text inside the card displaying the consultant's expertise/field.</i>
<i>BtnStart</i>	<i>Button</i>	<i>Start Chat</i>	<i>Button inside the card to initiate the session (Redirects to P00Y).</i>
<i>BtnMore</i>	<i>Button</i>	<i>Down Arrow</i>	<i>Show more of the consultant list.</i>

Page CHAT ROOM / P00Y

Logo

Home
Search
Profile

Chat dengan Mr. John Doe

Return

End

LblChatName

BtnReturn

BtnEnd

AreaContent

InputMessage

Type a new message here

+

📎

📌

🔍

1

2

3

4

5

6

7

8

9

10

1. BtnAdd

2. BtnSchedule

3. BtnPin

4. BtnReport

5. ListPeople

...

BtnMore

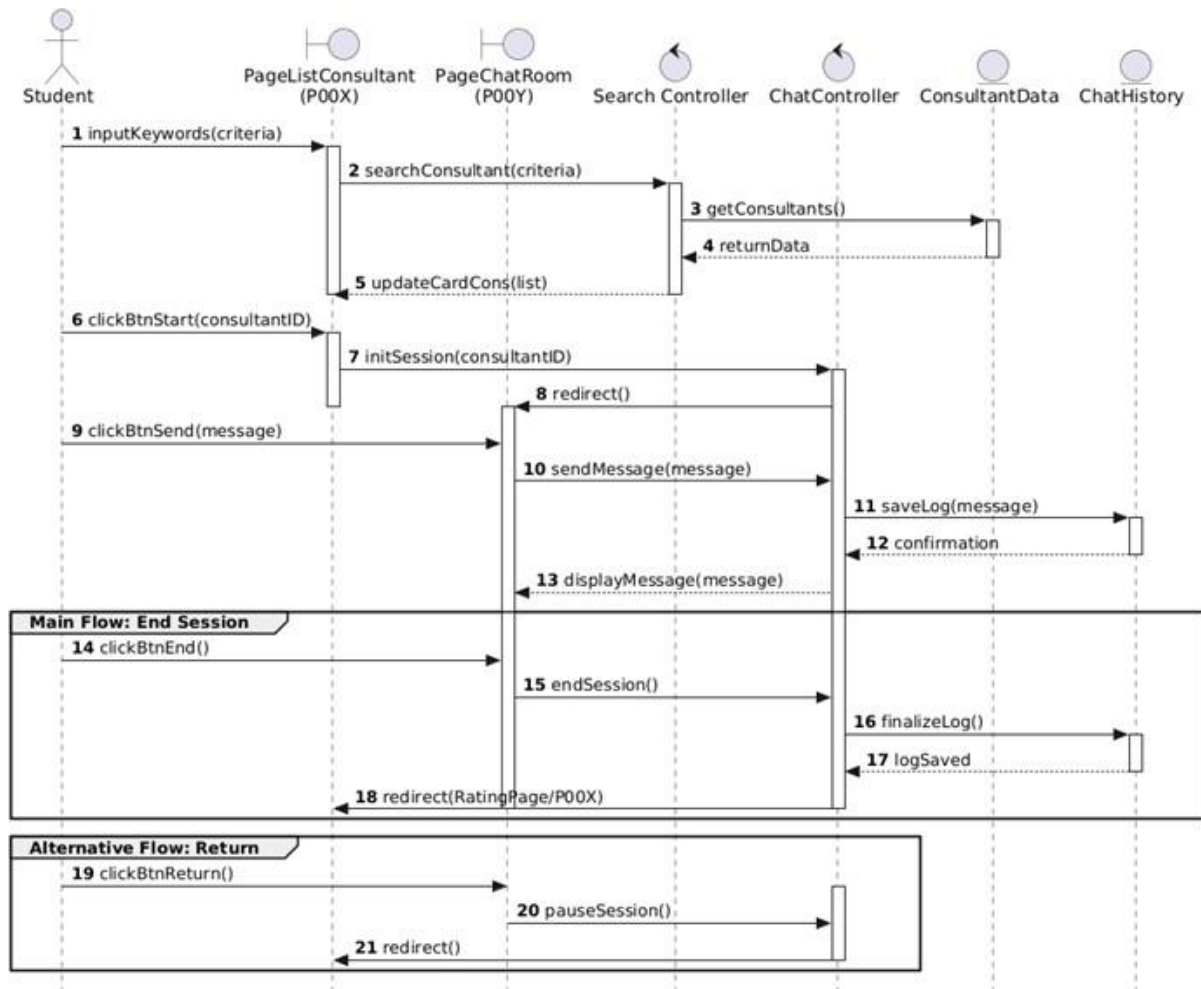
<i>Object_ID</i>	<i>TYPE</i>	<i>LABEL*</i>	<i>Description**</i>
<i>LblChatName</i>	<i>Label</i>	<i>Chat dengan: [Name]</i>	<i>Header label displaying the consultant's name.</i>
<i>BtnReturn</i>	<i>Button</i>	<i>Return</i>	<i>Navigates back to the list without ending the session.</i>
<i>BtnEnd</i>	<i>Button</i>	<i>End</i>	<i>Formally terminates the consulting session.</i>
<i>AreaContent</i>	<i>Dynamic List</i>	<i>(Chat History)</i>	<i>The main display area showing the history of message bubbles.</i>
<i>InputMessage</i>	<i>Input Field</i>	<i>(Type a message...)</i>	<i>The text field where the user types their message.</i>
<i>BtnSend</i>	<i>Button</i>	<i>(Send Icon)</i>	<i>Button to send the text in InputMessage to the consultant.</i>
<i>Btn.Add</i>	<i>Button</i>	<i>+</i>	<i>Icon to attach files or images.</i>

<i>BtnSchedule</i>	<i>Button</i>	<i>(Calendar Icon)</i>	<i>Icon to open the appointment scheduling modal.</i>
<i>BtnReport</i>	<i>Button</i>	<i>Flag</i>	<i>Button to report consultant for misconduct.</i>
<i>BtnPin</i>	<i>Button</i>	<i>Pin Mark</i>	<i>Button to pin/save current consultant.</i>
<i>ListPeople</i>	<i>Dynamic List</i>	<i>Person Outline</i>	<i>Icon to show how many people are in the chat.</i>
<i>BtnMore</i>	<i>Button</i>	<i>Three Dots</i>	<i>Button to show more settings option.</i>

3.1.3.3 Identification of Objects and Class Types #3 <Live chat consulting session with professionals>

No	Object Name	Object Type	Description
1	PageListConsultant (P00X)	Boundary (Interface)	The user interface for searching and selecting consultants.
2	PageChatRoom (P00Y)	Boundary (Interface)	The user interface for the live conversation and file sharing.
3	ConsultantData	Entity (Database)	Stores the profiles (Name, Photo, Speciality) displayed on CardCons in P00X.
4	ChatHistory	Entity (Database)	Stores the log of messages displayed in AreaContent in P00Y.
5	SearchController	Controller (Processing)	Handles the logic when the user interacts with InputSearch or ListFilter.
6	ChatController	Controller (Processing)	Handles message transmission (BtnSend), file uploads, and session termination (BtnEnd).

3.1.3.4 Sequence Diagram #3 <Live chat consulting session with professionals>



3.1.4 Use Case #4 <Give Review and Rating >

3.1.4.1 Use Case Scenario #4 <Give Review and Rating>

Skenario Use Case #4 :

- i. Pre-Condition : The user is aware of the quality of the desired university and can filter between general reviews or personal reviews.
- ii. Use Case Description : The user can provide a personal review or check the existing review and rating scores for every university.
 - c. Primary Flow
 9. **User** clicks on the university details to see more information.
 10. **User** sees the university profile along with its current reviews and ratings.
 11. **User** checks and reads reviews and ratings submitted by other users.
 12. **User** chooses to give their own review and rating.
 13. **User** sees a review and rating form, fills it out, and submits it to be shared across the platform.
 - d. Alternative Flow
 4. **User** views the ratings but decides not to provide a review or rating themselves.
 - 5.
- iii. Post-Condition : The user knows what to expect from the desired university based on existing feedback from others.

3.1.4.2 UI Design dan Deskripsi Objek UI #1 <Give Review and Rating>

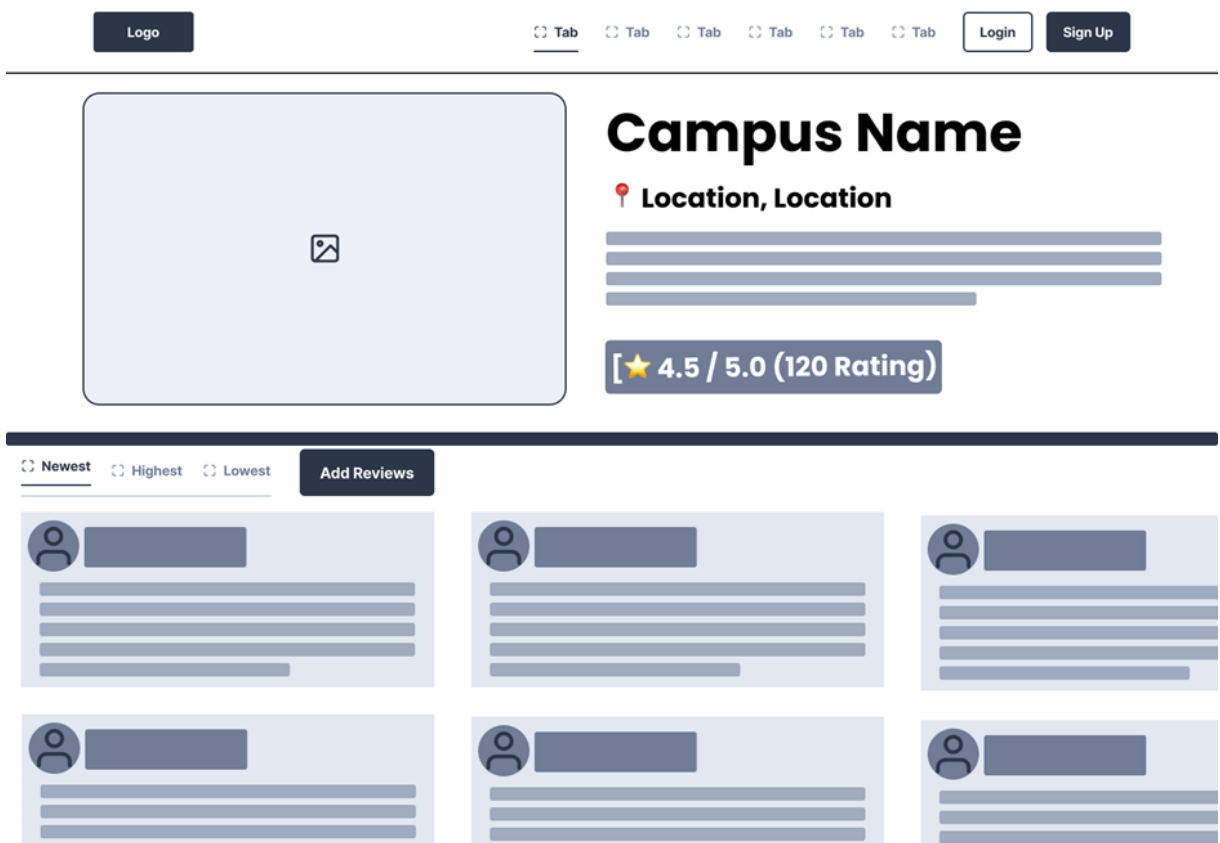
Bagian ini diisi dengan versi awal prototipe antarmuka untuk per Use Case
Selanjutnya, untuk setiap antarmuka/layar, tuliskan spesifikasi detilnya.

Tabel Deskripsi Objek UI

ID. PAGE	NAMA PAGE	DESCRIPTION
GRR01	<i>University Detail Page</i>	<i>The main page displays campus details, average ratings, and a list of reviews from other users.</i>
GRR02	<i>Form Review Modal</i>	<i>A pop-up (modal) display that appears when a user wants to give a rating and write a review.</i>

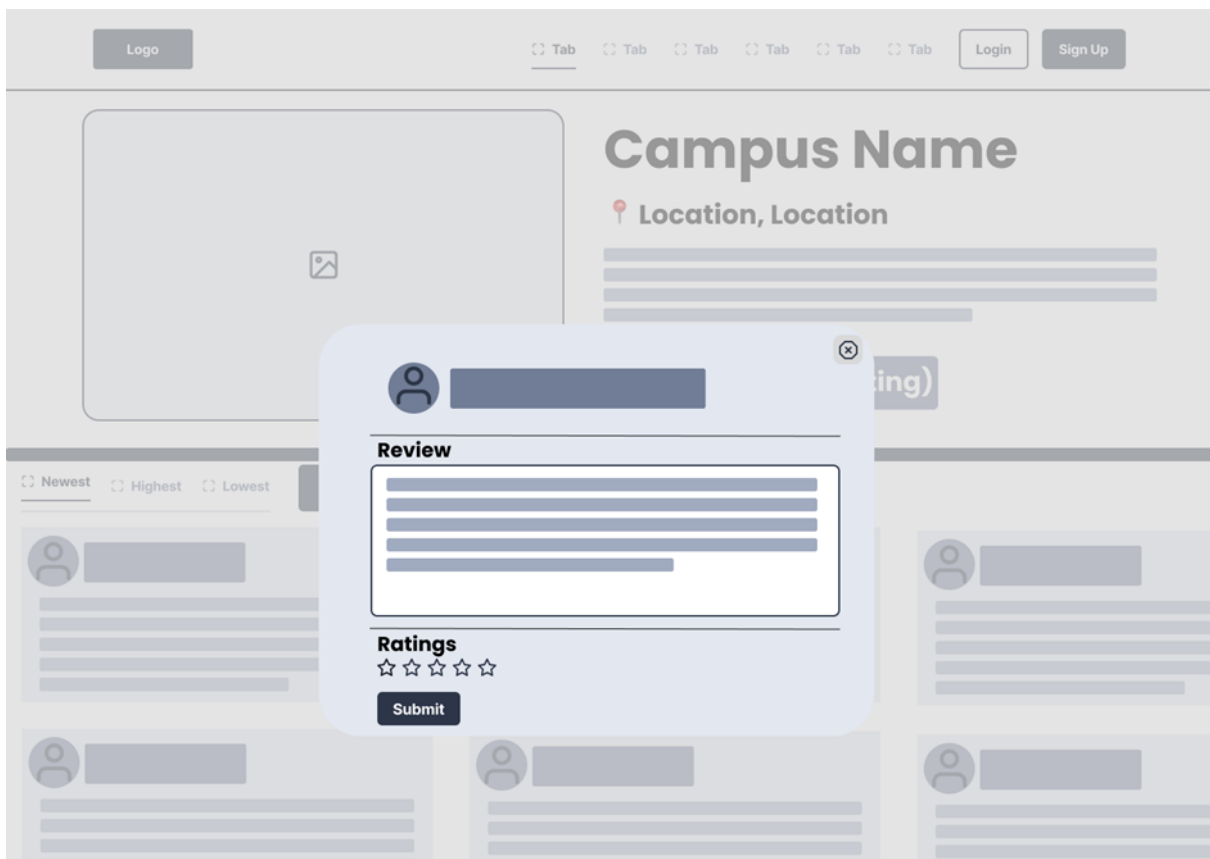
UNTUK MASING – MASING ANTAR MUKA / PAGE dibuatkan spesifikasi detil
Antarmuka XXX: {diisi dengan no. layar atau no gambar rancangan antarmuka}

GR001: University Detail Page



<u><i>Id Objek</i></u>	<u><i>JENIS</i></u>	<u><i>LABEL</i></u>	<u><i>Information</i></u>
<u><i>Img_Campus</i></u>	<u><i>Image</i></u>	=	<u><i>Header/main photo of the university.Header/main photo of the university.</i></u>
<u><i>Lbl_Name</i></u>	<u><i>Label</i></u>	<u><i>Campus Name</i></u>	<u><i>Displays the name of the selected university.</i></u>
<u><i>Lbl_AvgRate</i></u>	<u><i>Label</i></u>	<u><i>4.5 / 5.0 (120 Rating)</i></u>	<u><i>Displays the average rating calculation and total number of reviews.</i></u>
<u><i>Btn_Add</i></u>	<u><i>Button</i></u>	<u><i>Add Reviews</i></u>	<u><i>Lock Button: If clicked, will call/open page P002 (Review Form).</i></u>
<u><i>Btn_Filter</i></u>	<u><i>Button/Tab</i></u>	<u><i>Newest, Highest, Lowest</i></u>	<u><i>Sort the list of reviews based on specific criteria.</i></u>
<u><i>List_Review</i></u>	<u><i>Container</i></u>	=	<u><i>The area that displays a list of review cards from other users (Avatar, Name, Text, Stars).</i></u>

GR002: Form Review Model



<u><i>Id Objek</i></u>	<u><i>JENIS</i></u>	<u><i>LABEL</i></u>	<u><i>Information</i></u>
<u><i>Btn_Close</i></u>	<u><i>Button</i></u>	<u><i>(Icon X)</i></u>	<u><i>Closes the pop-up form without saving the data.</i></u>
<u><i>Inp_Rating</i></u>	<u><i>Input (Radio)</i></u>	<u><i>(Icon Bintang)</i></u>	<u><i>Input for user to choose score 1-5 stars.</i></u>
<u><i>Inp_Text</i></u>	<u><i>Text Area</i></u>	<u><i>Review</i></u>	<u><i>Text input box for user to write review/comment details.</i></u>
<u><i>Lbl User</i></u>	<u><i>Label</i></u>	<u><i>(User Name)</i></u>	<u><i>Displays the name of the currently logged in user.</i></u>
<u><i>Btn_Submit</i></u>	<u><i>Button</i></u>	<u><i>Submit / Post</i></u>	<u><i>Lock Button: If clicked, the system will Save the review data to the database and close the form.</i></u>

3.1.4.3 Identifikasi Object dan Tipe Kelas #1 <Give Review and Rating>

Identifikasi object yang terkait dengan use case tersebut.

Kelas pada tahap perancangan berbeda dengan kelas pada tahap analisis.

Gunakan tabel di bawah:

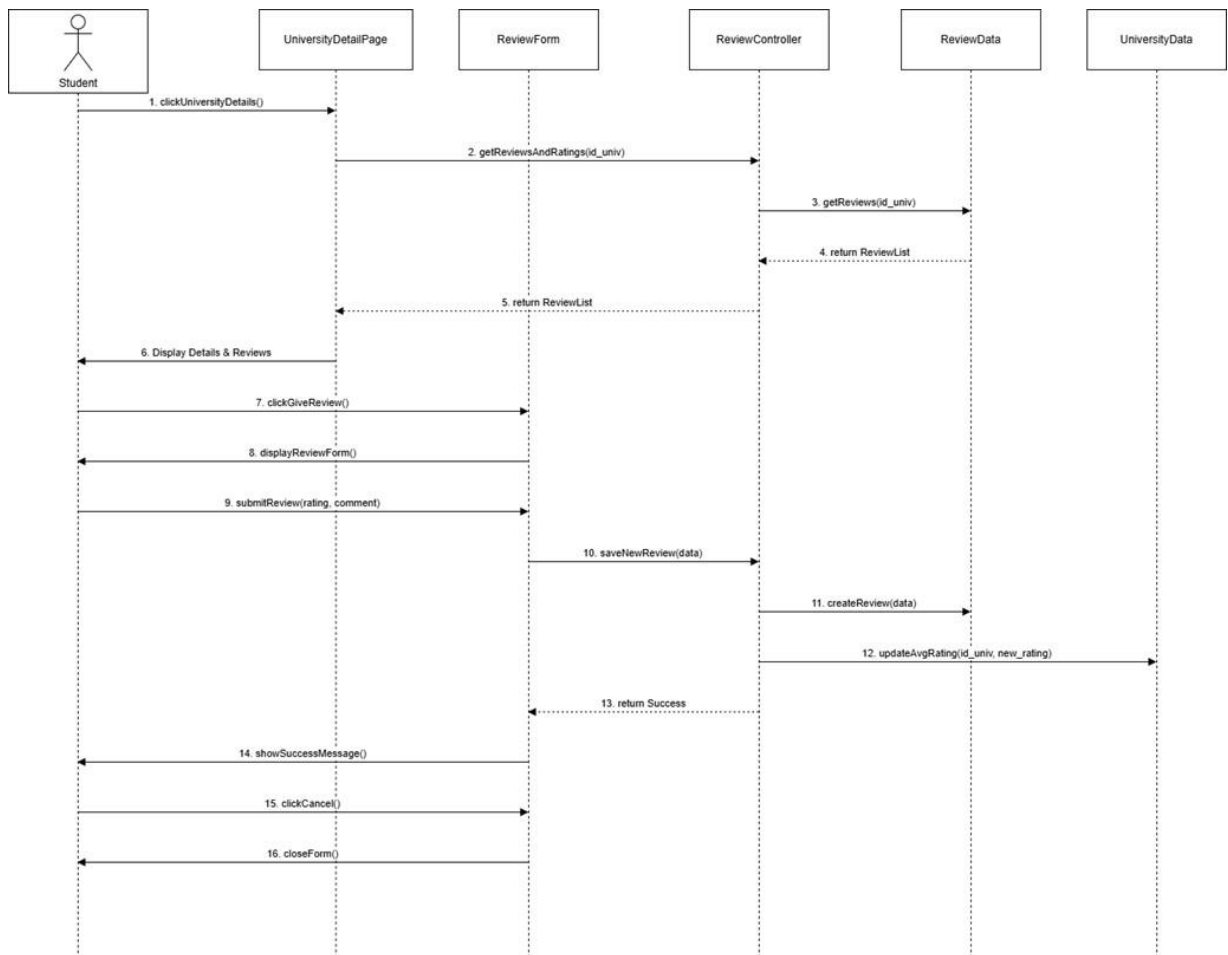
OBJECT TABLE: LOGIN

<i>No</i>	<i>Nama Object</i>	<i>Object Type</i>
1	UniversityDetailPage	Boundary (Detail Page Interface/UI)
2	ReviewForm	Boundary (Input Form Interface/UI)
3	ReviewController	Control (Logic that controls review storage & average calculation)
4	UniversityData	Entity (Campus data: Name, Location, Photo)
5	ReviewData	Entity (Review data: Star Score, Comment Content, Date)
6	UserData	Entity (User data: Name, Avatar who gave the review)

3.1.4.4 Sequence Diagram #4 <Give Review and Rating>

Buatlah diagram sequence untuk masing - masing use case sesuai skenario.

Skenario harus melibatkan kelas-kelas perancangan yang baru diidentifikasi.



3.1.5 Use Case #5 <Consult with AI Chatbot>

3.1.5.1 Use Case Scenario #5 <Consult with AI Chatbot>

Skenario Use Case #1 :

i. Pre-Condition : The user fully understands the ethics of AI usage, especially within the education sector.

ii. Use Case Description : The user can have a live consulting session with an AI to get insights and answers to their questions.

e. Primary Flow

1. **User** clicks the "AI Chatbot" option.
2. **User** sees a greeting from the AI asking if there is anything they want to ask.
3. **User** asks questions via text or voice note.
4. **User** receives answers based on their questions, including available data and descriptive explanations.
5. **User** provides extra questions regarding education as needed.
6. **User** continues to receive answers from the AI until they are satisfied.

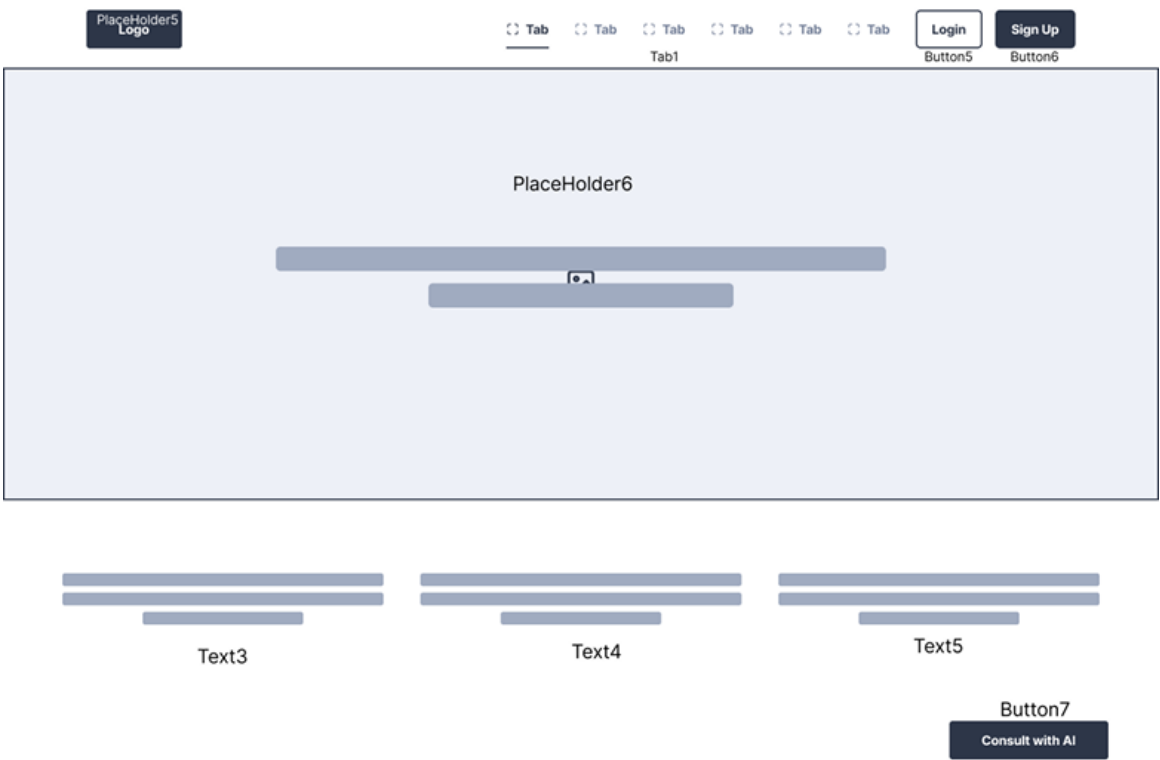
f. Alternative Flow

1. **User** asks about a university the AI cannot find, and sees an apologizing message displayed by the system.

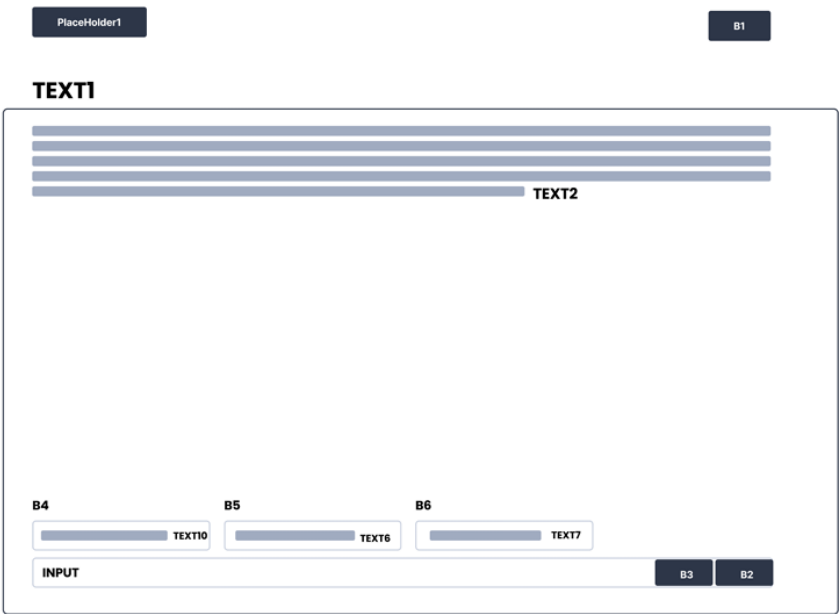
iii. Post-Condition : The user knows what to expect from the desired university based on existing feedback from others.

3.1.5.2 UI Design and UI Object Description #5 <Consult with AI Chatbot>

PAGE-001



PAGE-002



ID. PAGE	PAGE NAME	DESCRIPTION
PAGE-001	Home Page	Main page of the software which consist a button to go to the AI Chat Page.
PAGE-002	AI Chat Page	Page where user can consult with AI Chatbot.

PAGE-001 : Home Page

Object_Id	Type	Label*	Description
Placeholder5	Image	Logo	Placeholder for brand logo showing on the home page
Placeholder6	Content Area	Back to Home Page	Placeholder for the primary headline and sub-text (2 Lines Rectangle) in the main content area.
Text3	Text	Feature card 1	Placeholder to highlight one of the 3 features available
Text4	Text	Feature card 2	Placeholder to highlight the second feature
Text5	Text	Feature card 3	Placeholder to highlight the third feature
Tab1	Tab Menu	Tab 1-6	When clicked, the user will be directed to other content pages available
Button5	Button	Login	When clicked, the user will be directed to the login page.
Button6	Button	Sign Up	When clicked, the user will be directed to the Sign Up page.
Button7	Button	Consult with AI	When clicked, the user will be directed to the consultation with AI page (PAGE-002).

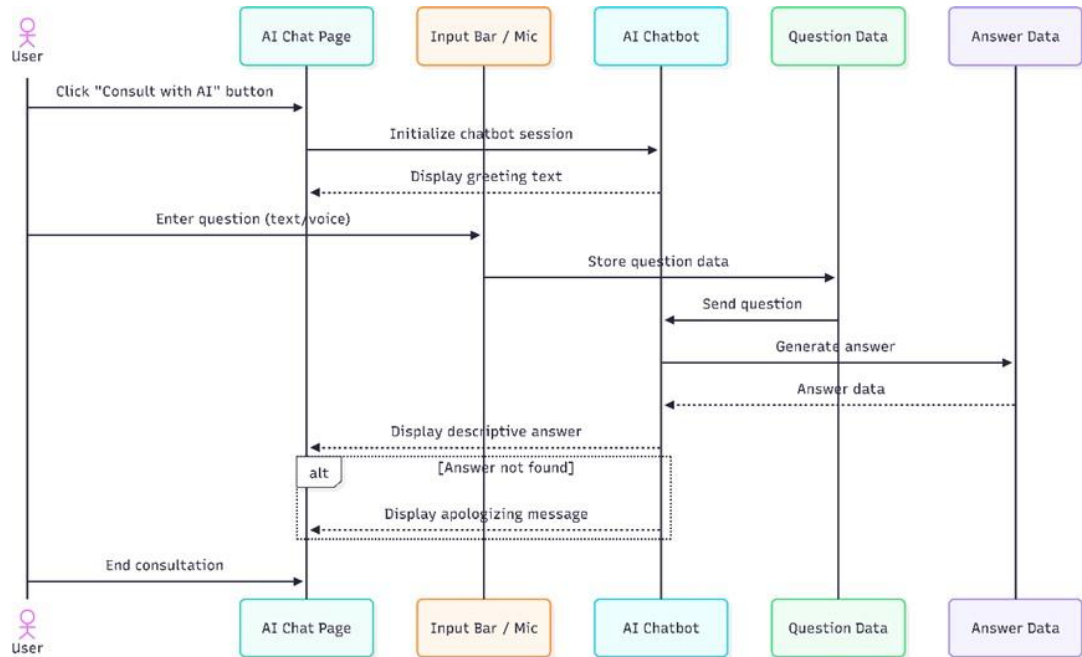
PAGE-002 : AI Chat Page

Object_Id	Type	Label*	Description
Placeholder1	Image	Logo	Interlect Logo
B1	Button	Home	When clicked, back to Home Page
TEXT1	Text	Page Title	“Consult With AI”
TEXT2	Text	Greeting Text	Greeting text from AI
TEXT10	Text	Most Asked Question	Display most asked question text
TEXT6	Text	Most Asked Question	Display most asked question text
TEXT7	Text	Most Asked Question	Display most asked question text
B2	Button	Up arrow symbol	When clicked, submit the question
B3	Button	Mic symbol	When clicked, turn on the voice search
B4	Button	TEXT10	When clicked, paste the question to input bar
B5	Button	TEXT6	When clicked, paste the question to input bar
B6	Button	TEXT7	When clicked, paste the question to input bar
INPUT	Input Field	“Tell me what you’d like to know!”	When clicked, the user will be able to input their question

3.1.5.4 Identifikasi Object dan Tipe Kelas #5 <Consult with AI Chatbot>

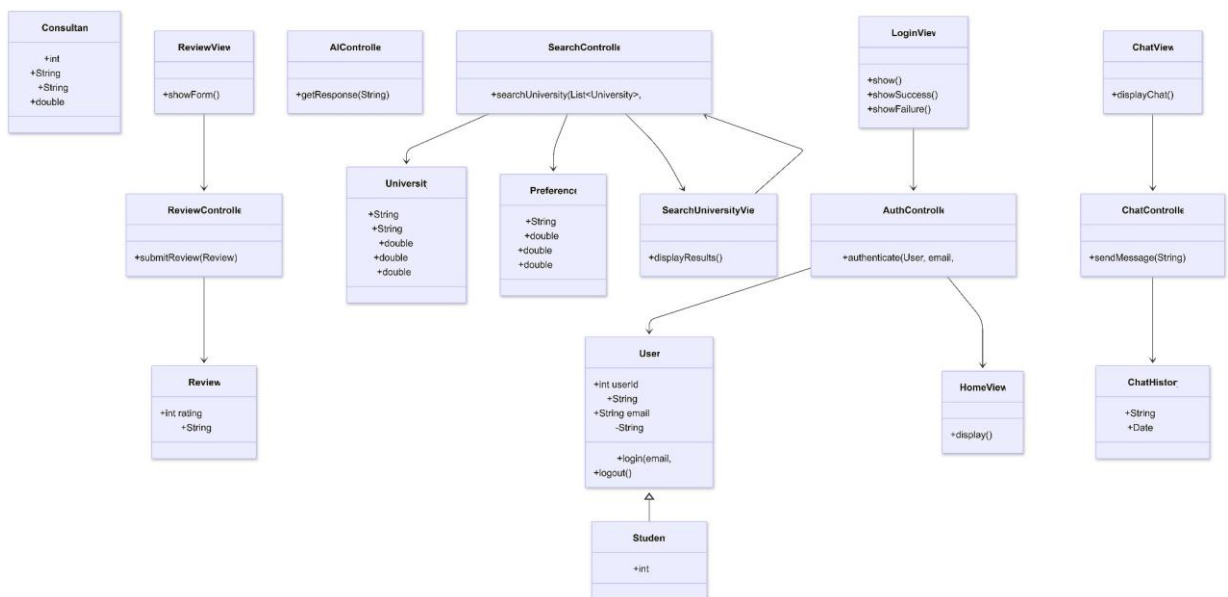
No	Object Name	Object Type
1	AI Chat Page (PAGE-002)	Boundary
2	Input bar	Boundary
3	Mic	Boundary
4	AI Chatbot	Controller
5	Greeting Text	Boundary
6	Question Data	Entity
7	Answer Data	Entity
8	Consult with AI Button	Boundary

3.1.5.5 Sequence Diagram #5 <Consult with AI Chatbot>



3.2 Overall Class Diagram

Bagian ini diisi dengan diagram kelas keseluruhan yang akan digunakan dalam PL menggunakan model MVC



3.3 Detailed Class Design

Bagian ini diisi dengan daftar keseluruhan kelas yang akan digunakan dalam PL menggunakan model MVC

CLASS TABLE :

ID Kelas	Nama Kelas Perancangan	Attribute (visibility)	Method / Operation
C01	User	+int userId +String name +String email - String password	+login(email, password) + logout()
C02	Student	+int studentID	(Inherits from User)
C03	SearchController		+searchUniversity(List<University>)
C04	AuthController		+authenticate(User, email, password)
C05	ReviewController		+submitReview(Review)
C06	ChatController		+sendMessage(String)
C07	AIController		+getResponse(String)
C08	University	+String name +String location +double rating +double tuition	
C09	Review	+int rating +String comment	
C10	ChatHistory	+String message +Date timestamp	
C11	Preference	+String category +double weight +double value	
C12	Consultant	+int id +String name +String expertise double rating	

3.4 Perancangan Algoritma dan/atau Query**Algoritma #1**

Nama Kelas : SearchScholarshipController

Nama Operasi : searchUniversity

Algoritma :

```

#include <iostream>
#include <vector>
#include <string>

struct University {
    std::string name;
    std::string country;
    std::string major;
    double livingAndTuitionCost;
    double minGPARequirement;
    double minIELTSRequirement;
};

struct Preference {
    std::string targetCountry;
    std::string desiredMajor;
    double maxBudget;
    double currentGPA;
    double ieltsScore;
};

class UniversitySearchController {
public:
    std::vector<University> searchUniversity(const std::vector<University>& database, const
Preference& userPref) {
        std::vector<University> matchedUniversities;

        for (const auto& uni : database) {
            bool isMatch = true;

            if (!userPref.targetCountry.empty() && uni.country != userPref.targetCountry) {
                isMatch = false;
            }
            if (isMatch && !userPref.desiredMajor.empty() && uni.major != userPref.desiredMajor) {
                isMatch = false;
            }
            if (isMatch && uni.livingAndTuitionCost > userPref.maxBudget) {
                isMatch = false;
            }
            if (isMatch && (userPref.currentGPA < uni.minGPARequirement)) {
                isMatch = false;
            }
            if (isMatch && (userPref.ieltsScore < uni.minIELTSRequirement)) {
                isMatch = false;
            }

            if (isMatch) {
                matchedUniversities.push_back(uni);
            }
        }
        return matchedUniversities;
    }
};

int main() {
    std::vector<University> universityDB = {
        {"Monash University", "Australia", "Computer Science", 25000.0, 3.0, 6.5},
        {"Oxford University", "UK", "Computer Science", 50000.0, 3.8, 7.5},
    };
}

```

```

        {"Telkom University", "Indonesia", "Informatics", 1000.0, 2.5, 5.0},
        {"University of Melbourne", "Australia", "Computer Science", 28000.0, 3.2, 7.0}
    };

    Preference studentPref;
    studentPref.targetCountry = "Australia";
    studentPref.desiredMajor = "Computer Science";
    studentPref.maxBudget = 30000.0;
    studentPref.currentGPA = 3.5;
    studentPref.ieltsScore = 7.0;

    UniversitySearchController controller;
    std::vector<University> results = controller.searchUniversity(universityDB, studentPref);

    if (results.empty()) {
        std::cout << "No university found." << std::endl;
    } else {
        std::cout << "Found " << results.size() << " universities:" << std::endl;
        for (const auto& uni : results) {
            std::cout << "- " << uni.name << " ($" << uni.livingAndTuitionCost << ")" << std::endl;
        }
    }

    return 0;
}

```

Query #2

Query :
SELECT consultant_id, name, speciality, rating, photo_url
FROM Consultant
WHERE speciality LIKE '%Input_Keyword%'
ORDER BY rating DESC;
Description : Retrieves a list of consultants matching the user's search keyword (speciality), ordered by their rating.

Query #3

Query :
INSERT INTO ReviewData (university_id, student_id, rating_score, comment, created_at)
VALUES ('Selected_UniID', 'Current_StudentID', Input_Rating, 'Input_Review_Text', NOW());
Description : Inserts a new record into the review table containing the rating score and comment text.

4 Requirement Traceability Matrix

FR Code	Functional Requirement Name	Use Case Name

FR-001	The system allows users to authenticate via email/password or external accounts.	Use Case #1 Login
FR-002	The system allows students to search for universities based on interest, budget, and capability (GPA/IELTS).	Use Case #2 Search/Generate Universities
FR-003	The system allows users to initiate a live chat session with professional consultants.	Use Case #3 Live chat consulting session with professionals
FR-004	The system allows users to provide ratings and reviews for universities.	Use Case #4 Give Review and Rating
FR-005	The system provides an AI-based chatbot for educational consultation.	Use Case #5 Consult with AI Chatbot
FR-006	The system allows users to compare details of multiple universities.	Use Case #2 Search/Generate Universities (Scenario: Compare)